Number of ν_e appearance events

$$N = \varepsilon_{\text{eff}}\, P(\nu_\mu \to \nu_e; E_0, L)\, \sigma_{CC}(E_0)\, \Phi\, \text{POT}\, N_T \;\simeq\; 60.2\, \varepsilon_{\text{eff}}\, P \left[\frac{\sigma_{CC}}{10^{-38}\,\text{cm}^2}\right] \left[\frac{\Phi}{10^{-14}\,\text{cm}^{-2}\,\text{POT}^{-1}}\right] \left[\frac{\text{POT}}{10^{21}}\right] \left[\frac{M}{\text{kt}}\right]$$

The second form follows from writing $N_T = M\, N_A$ (one nucleon per atomic mass unit, $N_A = 6.022 \times 10^{23}\,\text{g}^{-1}$) and measuring each factor in the units in brackets: the constant is $10^{-38} \cdot 10^{-14} \cdot 10^{21} \cdot (10^9\,\text{g/kt} \cdot 6.022 \times 10^{23}\,\text{g}^{-1}) = 60.2$, and all units cancel.

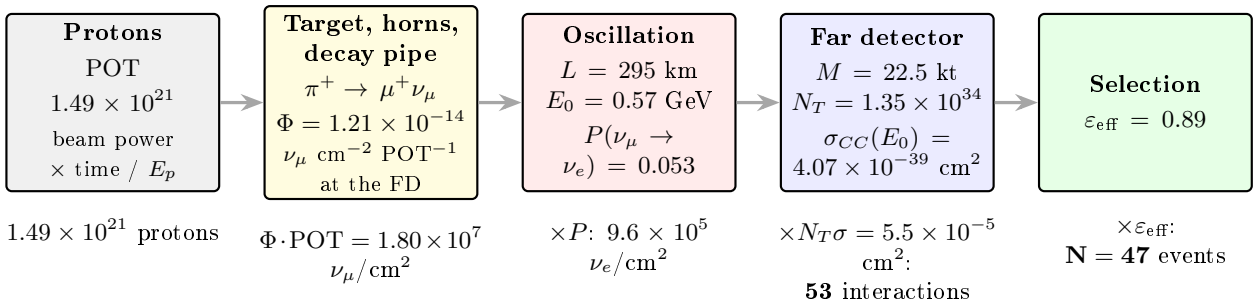


Figure 1: The chain of the calculation for the T2K neutrino beam (NuFIT 6.0, normal ordering). Each block is one factor of the formula; below each block, the running result. Protons on target produce, through the target, the focusing horns and the decay pipe, a ν_μ beam: at the far detector it gives Φ neutrinos per cm^2 per POT. A fraction P oscillates into ν_e over the baseline L; the far detector offers a target area $N_T\, \sigma_{CC}$; the selection keeps a fraction ε_{eff}.

- $\Phi \times \text{POT}$ is the **beam side**: the number of ν_μ crossing each cm^2 of the far detector. $\Phi = \int (d\Phi/dE)\, dE = \sum_k (d\Phi/dE)_k\, \Delta E_k$ is the ν_μ ($\bar{\nu}_\mu$) flux per proton on target *integrated over the whole energy spectrum* (not taken at E_0), summing the bins of the public flux histograms (T2K: Phys. Rev. D 87, 012001 (2013); NOvA: FERMILAB-DATA-2017-01; DUNE: arXiv:2103.04797), see Fig. 2. POT is the exposure, POT = beam power × time/proton energy.

- $\sigma_{CC} \times N_T$ is the **detector side**: the target area seen by a neutrino. $N_T = MN_A$ is the number of nucleons in the fiducial mass M; σ_{CC} is the inclusive charged-current cross section per nucleon (GLoBES tables, Comput. Phys. Commun. 167, 195 (2005); GENIE on argon for DUNE, arXiv:2103.04797), see Fig. 3.

- P is the oscillation probability in matter (NuFast, arXiv:2405.02400), here at the NuFIT 6.0 best fit (JHEP 12 (2024) 216). It is the only factor that depends on the oscillation parameters: varying δ_{CP}, the mass ordering or θ_{23} changes N only through P, all other factors fixed.

- E_0 replaces the energy spectrum by a single energy, the mean energy of the beam peak.

- ε_{eff} is an **effective parameter**. It accounts for everything the simple formula leaves out: detector efficiency and analysis cuts, the spread of the beam energy around E_0, and differences in cross sections. It is fixed so that the formula reproduces the signal predicted by each collaboration (T2K arXiv:2101.03779, NOvA arXiv:2509.04361, Hyper-K arXiv:1805.04163, DUNE arXiv:2103.04797). It is a ratio, not a detector efficiency: values above 1 (Hyper-K and DUNE $\bar{\nu}$ beams) come from the one-energy approximation, because the antineutrino cross section grows quickly with energy. With $\varepsilon_{\text{eff}} = 1$ the formula gives the number of CC interactions for a perfect detector (the choice of `tutorial_event_counting.ipynb`).

- **Assumptions**: pure ν_μ ($\bar{\nu}_\mu$) beam, one energy, no backgrounds, constant matter density. The totals agree with full simulations within ∼10 % for the ν beams and ∼30–40 % for the small $\bar{\nu}$ samples; the formula gives no energy spectrum.

Inputs (NuFIT 6.0 normal ordering for P).

	beam	L [km]	M [kt]	POT	E_0 [GeV]	Φ [10^{-14}/cm^2/POT]	σ_{CC} [10^{-38} cm^2]	ε_{eff}
T2K	ν	295	22.5	1.494×10^{21}	0.57	1.207	0.407	0.89
	$\bar{\nu}$			1.635×10^{21}	0.55	0.976	0.116	0.72
Hyper-K	ν	295	187	6.75×10^{21}	0.57	1.207	0.407	0.80
	$\bar{\nu}$			2.025×10^{22}	0.55	0.976	0.116	1.02
NOvA	ν	810	14	2.661×10^{21}	1.80	1.806	1.43	0.43
	$\bar{\nu}$			1.25×10^{21}	1.79	1.467	0.631	0.51
DUNE	ν	1300	40	7.15×10^{21}	2.31	1.548	1.78	0.62
	$\bar{\nu}$			7.15×10^{21}	2.27	1.385	0.729	1.07

Worked out ($N_{P=1} = 60.2\,\sigma\,\Phi\,\text{POT}\,M$ in the units above, then $\times P$, then $\times\varepsilon_{\text{eff}}$).

	beam	$N_{P=1}$	$N_{\varepsilon=1}$	N
T2K	ν	$60.2 \times 0.407 \times 1.207 \times 1.494 \times 22.5 = 994$	$\times 0.0533 = 53.0$	$\times 0.89 =$ **47**
	$\bar{\nu}$	$60.2 \times 0.116 \times 0.976 \times 1.635 \times 22.5 = 251$	$\times 0.0316 = 7.9$	$\times 0.72 =$ **5.7**
Hyper-K	ν	$60.2 \times 0.407 \times 1.207 \times 6.75 \times 187 = 37\,329$	$\times 0.0533 = 1990$	$\times 0.80 =$ **1592**
	$\bar{\nu}$	$60.2 \times 0.116 \times 0.976 \times 20.25 \times 187 = 25\,809$	$\times 0.0316 = 816$	$\times 1.02 =$ **832**
NOvA	ν	$60.2 \times 1.43 \times 1.806 \times 2.661 \times 14 = 5792$	$\times 0.0547 = 317$	$\times 0.43 =$ **136**
	$\bar{\nu}$	$60.2 \times 0.631 \times 1.467 \times 1.25 \times 14 = 975$	$\times 0.0249 = 24.3$	$\times 0.51 =$ **12.4**
DUNE	ν	$60.2 \times 1.78 \times 1.548 \times 7.15 \times 40 = 47\,441$	$\times 0.0738 = 3501$	$\times 0.62 =$ **2171**
	$\bar{\nu}$	$60.2 \times 0.729 \times 1.385 \times 7.15 \times 40 = 17\,384$	$\times 0.0178 = 309$	$\times 1.07 =$ **331**

Sensitivity to CP violation. Each experiment measures two numbers, N_ν and $N_{\bar{\nu}}$. The agreement between observed numbers N_i^{obs} and the prediction N_i^{pred} of a hypothesis is measured with

$$\chi^2 = \sum_{i=\nu,\bar{\nu}} \frac{\left(N_i^{\text{obs}} - N_i^{\text{pred}}\right)^2}{\sigma_i^2}, \qquad \sigma_i^2 = N_i^{\text{obs}},$$

since the number of events is Poisson distributed and its statistical variance equals the number of events (no normalization uncertainty is included). For the expected sensitivity: (i) the observed numbers are the prediction for the true value, $N_i^{\text{obs}} = N_i(\delta_{CP})$, without fluctuations, so the best fit is the true value itself, with $\chi^2 = 0$; (ii) CP conservation means $\delta_{CP} = 0$ or π: both must be rejected, so we keep the one that fits best; (iii) $\Delta\chi^2$, the difference from the best fit ($\chi^2 = 0$), is then

$$\boxed{\Delta\chi^2(\delta_{CP}) = \min_{\delta' \in \{0,\pi\}} \sum_{i=\nu,\bar{\nu}} \frac{\left[N_i(\delta_{CP}) - N_i(\delta')\right]^2}{N_i(\delta_{CP})}, \qquad \text{significance} = \sqrt{\Delta\chi^2}\,\sigma}$$

with N_i from the formula above (only P changes with δ_{CP}). For one parameter, $\Delta\chi^2 = 9$ (25) corresponds to 3σ (5σ) (PDG, Review of Particle Physics, Statistics chapter). The CP-violation sensitivity is defined, as in the DUNE (arXiv:2002.03005) and Hyper-K (arXiv:1805.04163) design studies, as $\sqrt{\Delta\chi^2}$ with respect to $\delta_{CP} = 0, \pi$ computed on Asimov data. Our version is a simplified counting version: Gaussian statistics, two samples, no backgrounds, fixed oscillation parameters. Derivation and the case with a normalization uncertainty: `docs/number_of_events.pdf`.

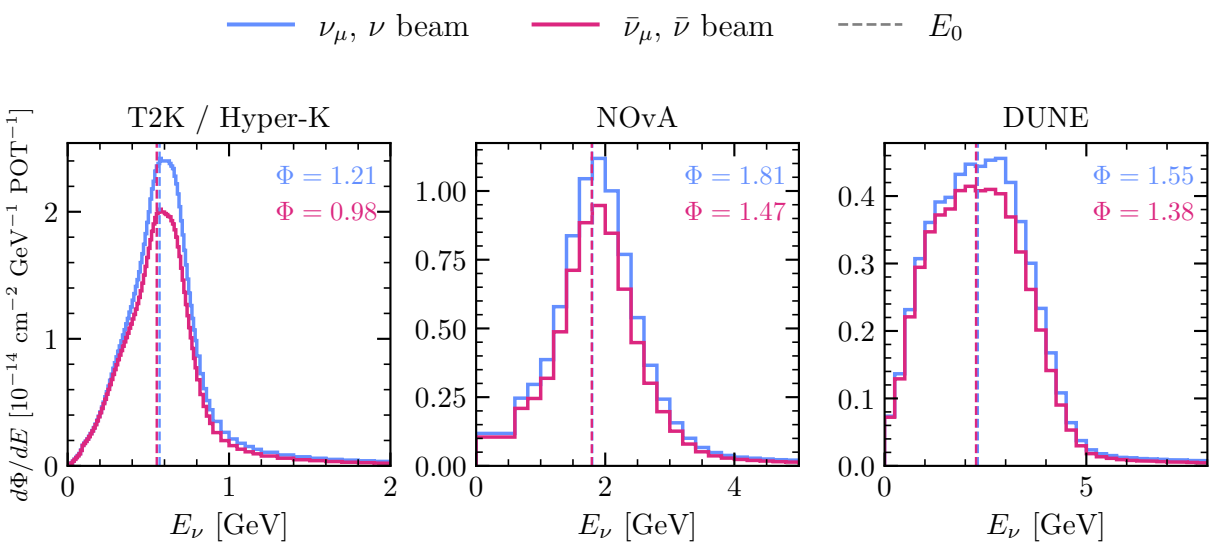


Figure 2: Unoscillated ν_μ flux in the neutrino beam and $\bar{\nu}_\mu$ flux in the antineutrino beam at the far detectors, per proton on target, from the public flux releases (T2K: 2020 release at Super-Kamiokande, also used for Hyper-K; NOvA: 2017 near-detector flux $\times (1\,\mathrm{km}/810\,\mathrm{km})^2$; DUNE: TDR far-detector flux). Dashed lines: the beam energy E_0 (flux-weighted mean energy of the main peak). The numbers give $\Phi = \sum_k (d\Phi/dE)_k \, \Delta E_k$ in units of 10^{-14} cm^{-2} POT^{-1}, as used in the table.

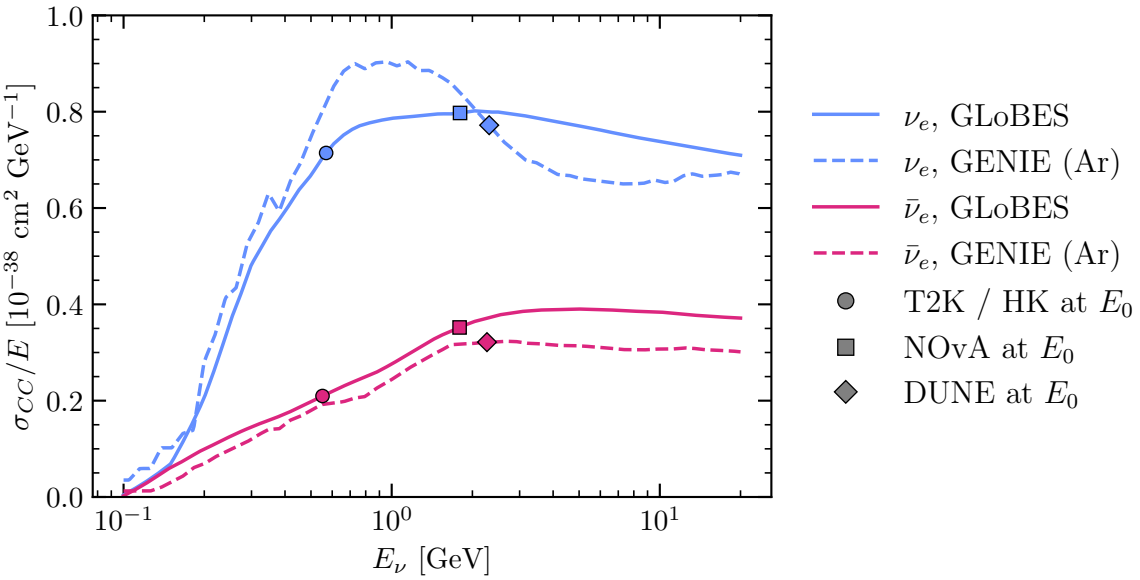


Figure 3: Inclusive charged-current cross section per nucleon, divided by energy, for ν_e and $\bar{\nu}_e$: GLoBES standard table (low energy: Super-Kamiokande cross sections for water, M. D. Messier, PhD thesis (1999); high energy: Paschos and Yu, Phys. Rev. D 65, 033002 (2002)), used for T2K, Hyper-K and NOvA; GENIE on argon of the DUNE TDR configuration (arXiv:2103.04797), used for DUNE. Markers: $\sigma_{CC}(E_0)/E_0$ entering the table.